

#This code will be used to demonstrate proposition 5.6. We demonstrate that the component curve $\tilde{n}k=0$ admits a single connected component are positive along the curve

Define nk

$$nk := (w^{-3} - x^{-3}) \cdot (y^{-3} - z^{-3}) + (z^{-3} - x^{-3}) \cdot (y^{-3} - w^{-3})$$

$$nk := \left(\frac{1}{w^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{z^3} \right) + \left(\frac{1}{z^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{w^3} \right) \quad (1)$$

#Apply the rational parametrization constructed in this chapter 4.

$$parametrization := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify(numer(expand(subs(parametrization, nk))));

simplify(denom(expand(subs(parametrization, nk))));

$$12 b^3 \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right.$$

$$+ \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6}$$

$$- \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2}$$

$$\left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right) a^3$$

$$(a^2 + 1)^3 (b + a)^3 (a b - 1)^3 (b^2 + 1)^3 (a b + 1)^3 (-b + a)^3 \quad (2)$$

Define ñk

$$\tilde{n}k := \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right.$$

$$+ \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6}$$

$$- \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2}$$

$$\left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right) :$$

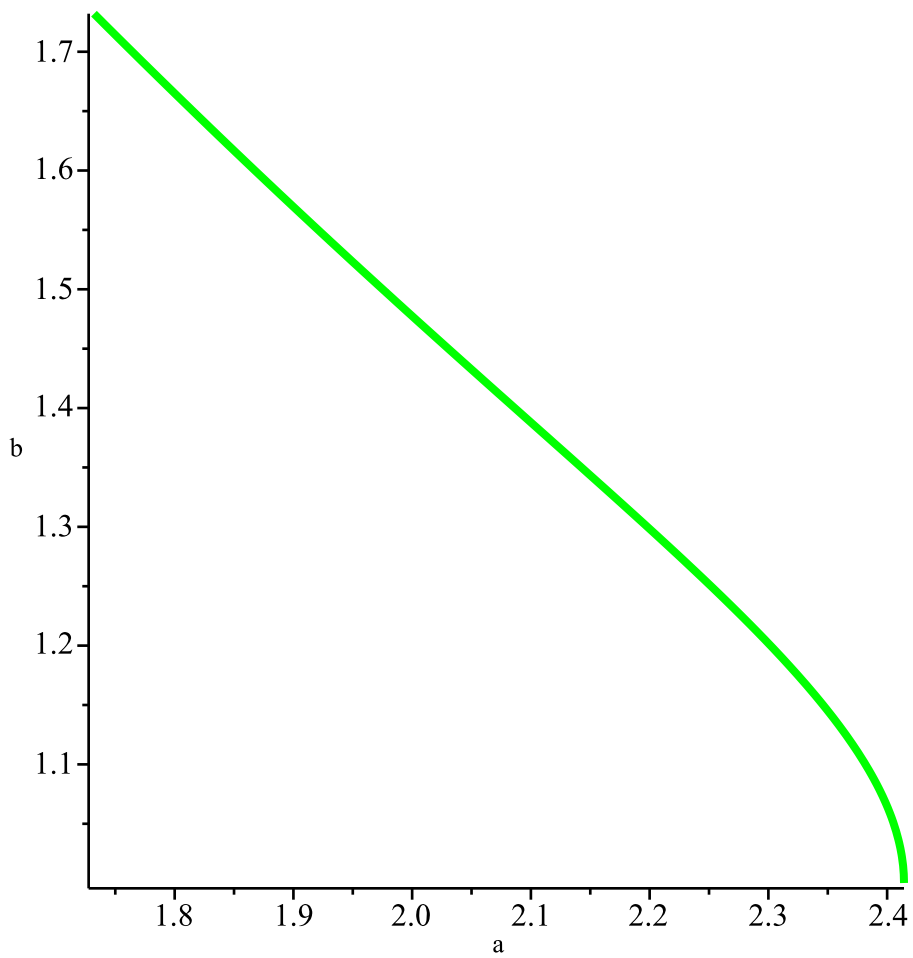
#Plot the curves $\tilde{n}k=0$ in region A1.

$r := a - b :$

with(plots) :

```
implicitplot([n̄k=0, r=0], a=sqrt(3) .. 1 + sqrt(2), b=1 .. sqrt(3),
  scaling=constrained,
  gridrefine=2,
  thickness=3,
  color=[green, red],
  labels=["a", "b"],
  title="Graphic for n̄k=0 in A1");
```

Graphic for $\tilde{n}k=0$ in A1



#Apply the Möbius transformation on the rectangle $A1 = [\text{sqrt}(3), 1 + \text{sqrt}(2)) \times [1, \text{sqrt}(3))$ constructing $n\tilde{k}$

$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{\text{sqrt}(3) + (1 + \text{sqrt}(2)) c}{1 + c}, \\ b = \frac{1 + \text{sqrt}(3) \cdot d}{1 + d} \end{array} \right\} :$$

$n\tilde{k} := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, \tilde{n}k))), c) :$

#Compute the degree of $n\tilde{k}$ in c

$\text{degree}(n\tilde{k}, c);$

10

(3)

#Define the coefficients $c_i = \text{coeff}(n\tilde{k}, c, i)$

$\text{seq}(\text{evalf}(\text{assign}(\text{cat}('c', i), \text{coeff}(n\tilde{k}, c, i))), i = 0 .. 10);$

#Compute the all degree of c_i in d

$\text{seq}(\text{degree}(\text{eval}(\text{cat}('c', i))), d, i = 0 .. 10);$

11, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12

(4)

#From here we will analyze the signal variation of all coefficients c_i

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c10, d, i))), i = 12 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(5)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c9, d, i))), i = 12 .. 0, -1);$

$\text{fsolve}(c9 = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1

0.3129734741

(6)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c8, d, i))), i = 12 .. 0, -1);$

$\text{fsolve}(c8 = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1

0.5479957779

(7)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c7, d, i))), i = 12 .. 0, -1);$

$\text{fsolve}(c7 = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1

0.8235234238

(8)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c6, d, i))), i = 12 .. 0, -1);$

$\text{fsolve}(c6 = 0, d, 0 .. \text{infinity});$

$$1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$1.176848249$$

(9)

seq(sign(evalf(coeff(c5, d, i))), i = 12 .. 0, -1);
fsolve(c5 = 0, d, 0 .. infinity);

$$1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$1.662545488$$

(10)

seq(sign(evalf(coeff(c4, d, i))), i = 12 .. 0, -1);
fsolve(c4 = 0, d, 0 .. infinity);

$$1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$2.384702176$$

(11)

seq(sign(evalf(coeff(c3, d, i))), i = 12 .. 0, -1);
fsolve(c3 = 0, d, 0 .. infinity);

$$1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$3.582998097$$

(12)

seq(sign(evalf(coeff(c2, d, i))), i = 12 .. 0, -1);
fsolve(c2 = 0, d, 0 .. infinity);

$$1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$5.973633668$$

(13)

seq(sign(evalf(coeff(c1, d, i))), i = 12 .. 0, -1);
fsolve(c1 = 0, d, 0 .. infinity);

$$1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$13.13379621$$

(14)

seq(sign(evalf(coeff(c0, d, i))), i = 11 .. 0, -1);

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

(15)

#This prove that the curve ñk is crossing vertically the rectangle A1

$$\begin{aligned} \tilde{\text{nk}} := & \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2b^6 + 8b^4 - 2b^2 + 1) a^8}{2} \right. \\ & + \frac{b (b^4 - 4b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8b^4 + 1) a^6}{6} \\ & - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2b^6 - 2b^4 + 2b^2 + 1) a^4}{2} \\ & \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right): \end{aligned}$$
$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{1 + \text{sqrt}(3) \cdot c}{1 + c}, \\ b = \frac{1 + \text{sqrt}(3) \cdot d}{1 + d} \end{array} \right\} :$$

```
#Compute the degree of  $nk$  in  $c$ 
```

10

(16)

```

for  $i$  from 0 to 10 do
   $c[i] := \text{coeff}(\text{'nk'}, c, i);$ 
od;

```

12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 11

(17)

$$seq(seq(sign(evalf(coeff(c[j], d, i))), i = 12..0, -1), j = 10..0, -1);$$
[illegible]

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$
 $-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$
 $-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$
 $-1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$\text{coeff}(c[10], d, 12) = 0 \tag{19}$$

#This prove that the curve ñk does not have solution in A2

```
restart
# Define nk
```

$$\begin{aligned} \tilde{\text{nk}} := & \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2b^6 + 8b^4 - 2b^2 + 1) a^8}{2} \right. \\ & + \frac{b (b^4 - 4b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8b^4 + 1) a^6}{6} \\ & - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2b^6 - 2b^4 + 2b^2 + 1) a^4}{2} \\ & \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right): \end{aligned}$$

#Apply the Möbius transformation on the rectangle $A3 = [1 + \text{sqrt}(2), \text{infinity}) \times [1, \text{sqrt}(3))$ constructing $n\mathbf{k}$

$$\text{moebius} := \left\{ \begin{array}{l} a = (1 + \text{sqrt}(2)) + c, \\ b = \frac{1 + \text{sqrt}(3) \cdot d}{1 + d} \end{array} \right\}:$$

$$\mathbf{nk} := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, \mathbf{\tilde{nk}}))), c) :$$

#Compute the degree of $n\mathbf{k}$ in c

$degree(n\hat{k}, c);$

10

(20)

#Define the coefficients $c_i = \text{coeff}(n\mathbf{k}, c, i)$

for i from 0 to 10 do
$$c[i] := \text{coeff}(\text{'nk'}, c, i);$$

od:

#Compute the all degree of ci in d

```
seq( degree( c[i], d ), i = 0 .. 10 );
```

12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12

(21)

#From here we will analyze the signal variation of all coefficients c_i

$$seq(seq(sign(evalf(coeff(c[j], d, i))), i = 12..0, -1), j = 10..0, -1);$$
[illegible]

[illegible]

#This prove that the curve $\tilde{n}_k$ does not have solution in A3

```
restart
# Define nk
```

$$\begin{aligned} \tilde{n}k := & \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \\ & + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\ & - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\ & \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right) : \end{aligned}$$

#Apply the Möbius transformation on the rectangle $A4 = [\sqrt{3}, \infty) \times [\sqrt{3}, \infty)$ constructing $n\bar{k}$

$$\text{moebius} := \{ a = \text{sqrt}(3) + c, \\ b = \text{sqrt}(3) + d \} :$$
$$\mathbf{nk} := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, \tilde{\mathbf{nk}}))), c) :$$

#Compute the degree of $n\mathbf{k}$ in c

 $degree(nk, c);$

10

(23)

#Define the coefficients $c_i = \text{coeff}(nk, c, i)$

for i from 0 to 10 do
$$c[i] := \text{coeff}(\text{'nk'}, c, i);$$

od:

#Compute the all degree of c_i in d

```
seq( degree(c[i], d), i = 0 .. 10 );
```

12, 12, 12, 12, 12, 12, 12, 11, 10, 8, 8

(24)

#From here we will analyze the signal variation of all coefficients c_i

$$seq(seq(sign(evalf(coeff(c[j], d, i))), i = 12..0, -1), j = 10..0, -1);$$
[illegible][illegible][illegible][illegible]
$$coeff(c[10], d, 12)$$

#This prove that the curve $\tilde{n}k$ does not have solution in A_4